EXP6

Write a program to implement Naïve Bayes algorithm in python and to display the results using confusion matrix and accuracy.

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CSV FILE

```
Outlook,Temperature,Humidity,Windy,Play
Sunny,Hot,High,False,No
Sunny,Hot,High,True,No
Overcast,Hot,High,False,Yes
Rain,Mild,High,False,Yes
Rain,Cool,Normal,False,Yes
Rain,Cool,Normal,True,No
Overcast,Cool,Normal,True,Yes
Sunny,Mild,High,False,No
Sunny,Cool,Normal,False,Yes
Rain,Mild,Normal,False,Yes
Sunny,Mild,Normal,True,Yes
Overcast,Mild,High,True,Yes
Overcast,Hot,Normal,False,Yes
Rain,Mild,High,True,No
```

PROGRAM

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OrdinalEncoder
from sklearn.naive_bayes import CategoricalNB
from sklearn.metrics import confusion_matrix, accuracy_score

data = pd.read_csv("weather.csv")

print(data)

X = data.iloc[:, :-1]
y = data.iloc[:, -1]

encoder = OrdinalEncoder()
X = encoder.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(
  X,
  y,
  test_size=0.3,
  random_state=42
)

model = CategoricalNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("\nActual Classes:")
```

print(y_test.values)

print("\nPredicted Classes:")

print(y_pred)

print("\nConfusion Matrix:")

print(confusion_matrix(y_test, y_pred))

print("\nAccuracy:")

print(accuracy_score(y_test, y_pred))

OUTPUT